Learn X+ Python Studio

Unleash Your Code!

Lesson 2 : Git & Github

Outline :

A) What is Git?

B) How Git Works?

C) What is Github?

D) How Github Complements Git

A) What is Git?

Version Control

Version control is a system that tracks changes to files over time, so that:

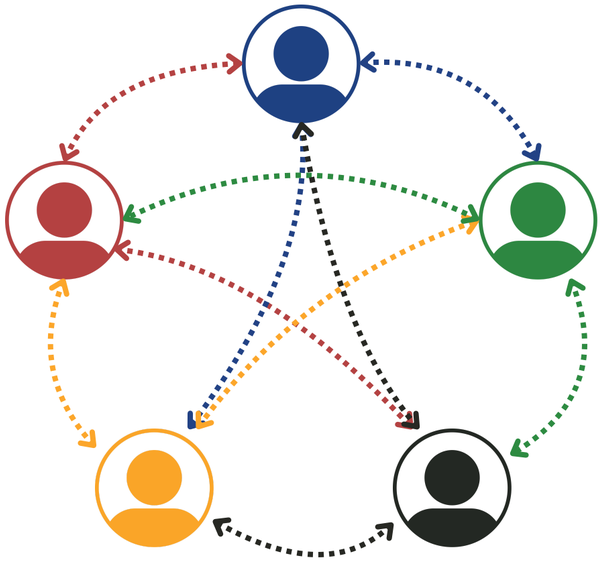
- See what was changed.
- See who changed it.
- Revert to previous state.
- Work on multiple versions simulataneously. (parallel)

A) What is Git?

Distributed

- **Everyone** has **original copy**.
- **Syncing** between collaborators.
- Everyone can **work independently.**

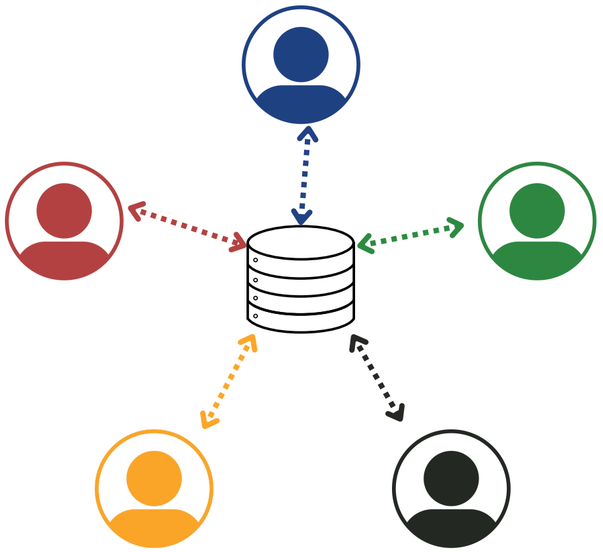
git mercurial



Centralised

- **One server** holds full history.
- **Everyone** checks in & out from it.
- If **server down,** no one can work.

SUBVERSION Google Docs



A) What is Git?

History

- **Created** by Finnish programmer named Linus Torvalds. (creater of Linux)
- **Developed** in April 2005.
- Created out of neccessity, Bitkeeper (version control) was no longer free.

- After 4 months, **lead maintainer** is Japanese Junio Hamano, from Google.
- Roughly **10-15 active maintainer**, with Junio reviewing & merging updates.

Linus Torvalds
source: Britannica

Junio Hamano
source: Github

B) How Git Works?

Under the Hood

Tree

- Directory structure.
- **Maps** blobs name.

Blobs

- File contents.
- No name, **just data.**

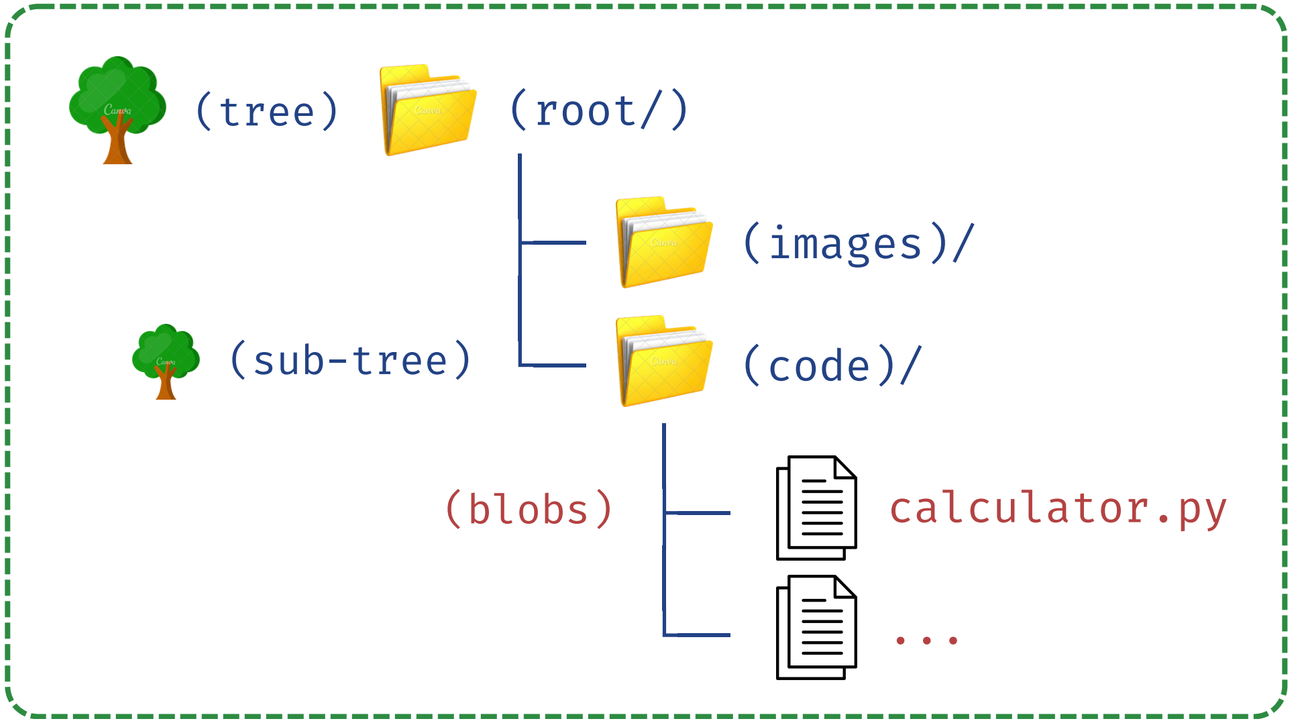
Commit

- Points to one root tree.
- **Author, timestamp, message.**

Tag/Hash

- Unique (no collision).
- **Identify commits.**

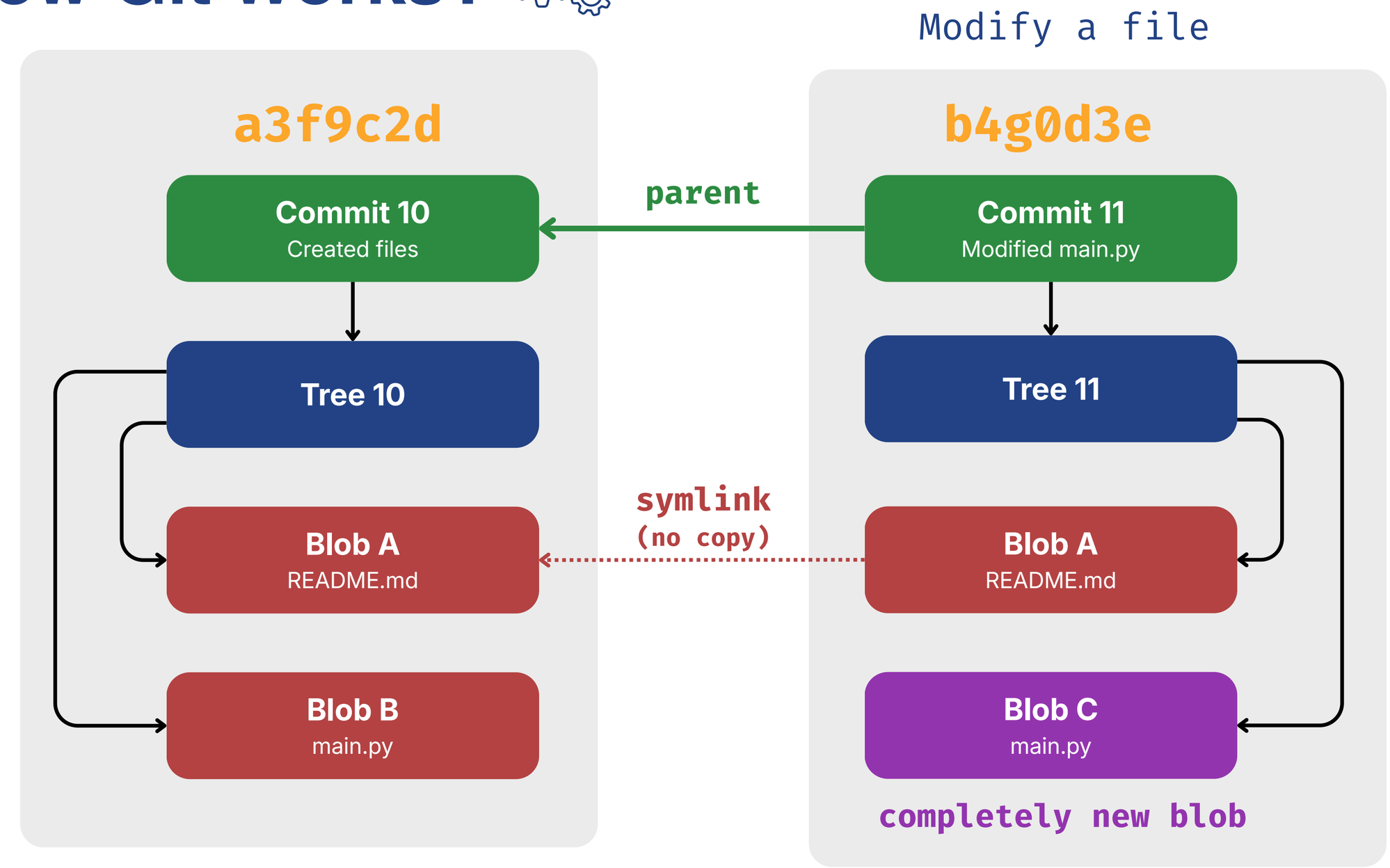
Commit 10 - a3f9c2d



"Added calculator .py"

Jim (29/2/2016)

B) How Git Works?

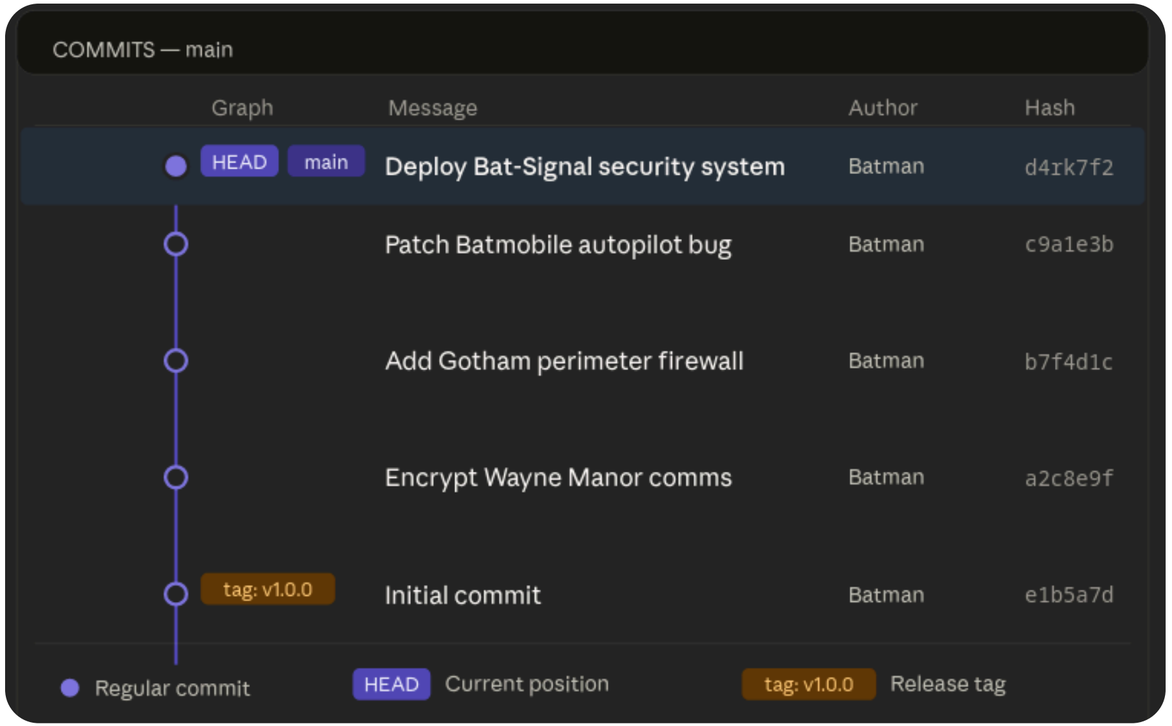


B) How Git Works?

Important Concepts

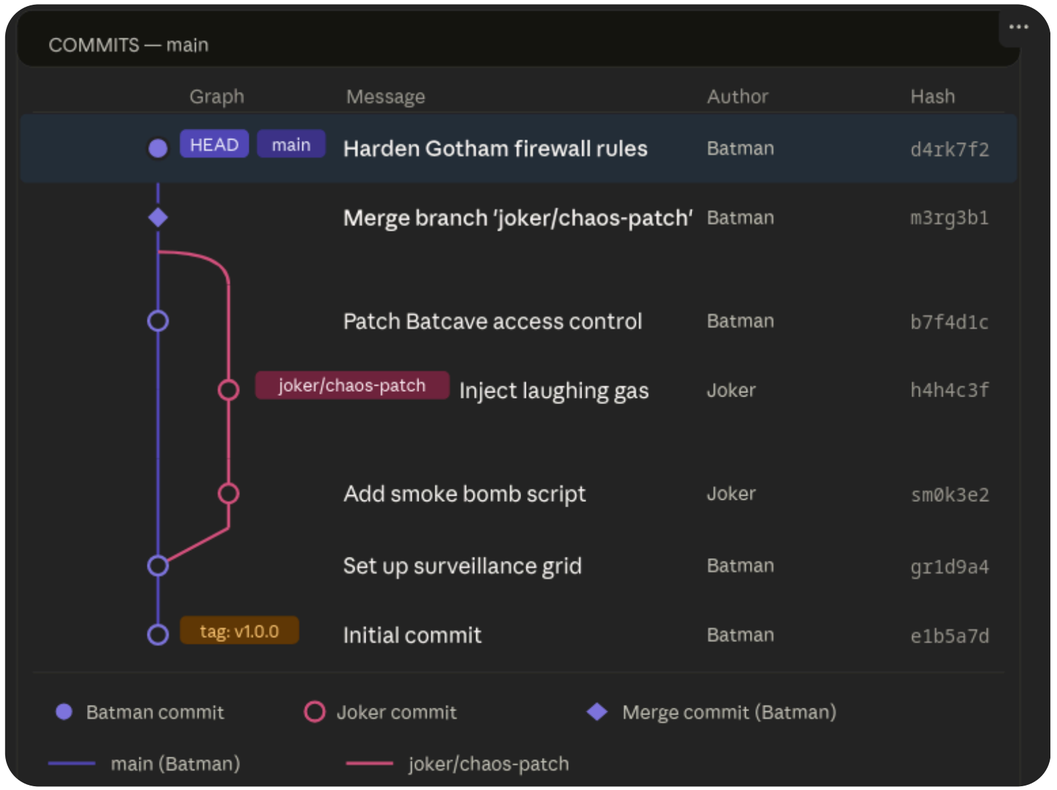
Commit

- Personal **snapshot** of project at a **specific point in time.**



Branch

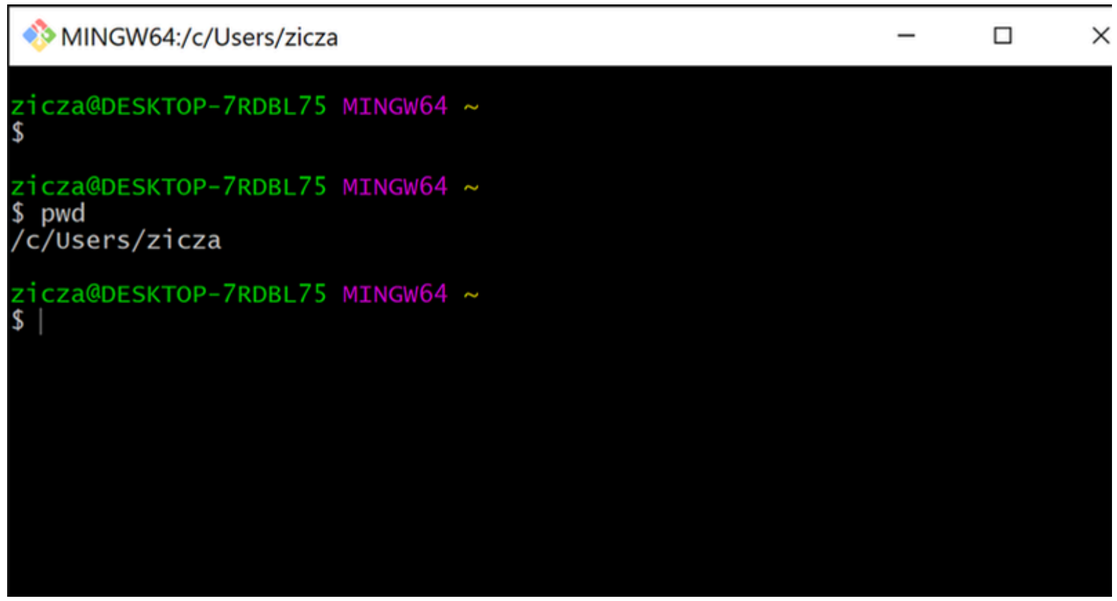
- **Independent section** so you can develop in parallel.



B) How Git Works?

Git Bash

- **Not real** Bash terminal.
- Windows application that **emulates Unix shell.**
- Only get **subset** of commands like:
  - ls
  - mkdir
  - cat
  - touch
- Unfortunately, missing man pages.



Git Bash shell

C) What is Github?

History

- **Founded** in April 2008.
- **Web-based hosting service** for Git repositories.

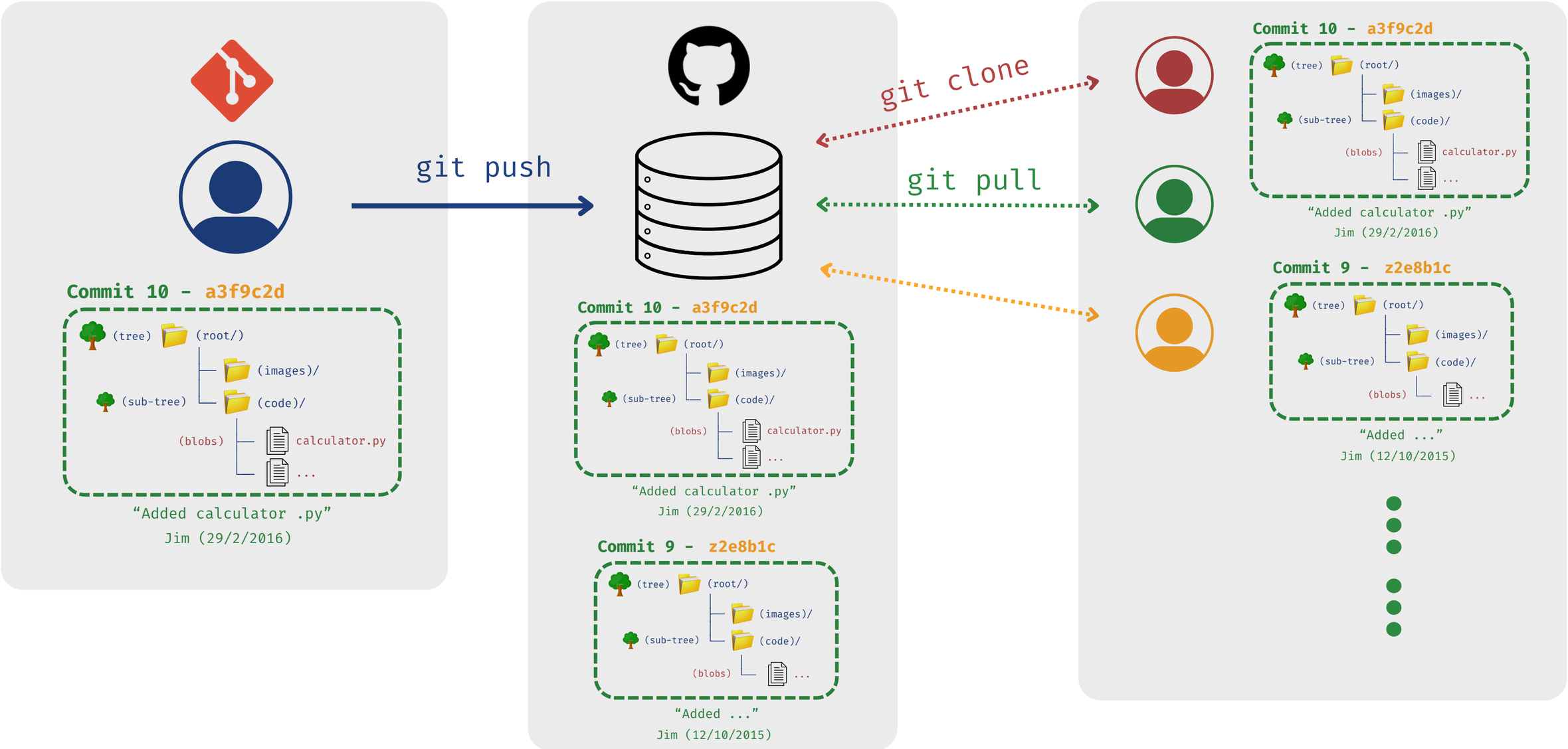
- **Acquired** by Microsoft in 2018 for $7.5 billion.
- **Uses your** public repo to train Copilot.
- **1 billion** repositories as of 2025.

P.J. Hyett (left), Tom Preston-Werner (center) and Chris Wanstrath.

https://www.inc.com/30under30/christine-lagorio/github-pj-hyett-chris-wanstrath-2013.html

D) How Github Complements Git

How Github Works



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End of Lesson 2